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Economics A: Microeconomics I, Group 2

Fall Semester 2026

Part 1.1

First Principles

Dr. Magnus Hoffmann



University of St.Gallen

Microeconomics I: Where We Are

Part 1: Introduction

Part 1.1: First Principles

Part 1.2: Gains from Trade

Part 2: A Primer in Markets and Institutions

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4. Theories and Models



Learning Goals: this part covers the following topics

- ▶ Enough philosophy of science to be able to have a qualified opinion about the status and scientific role of economic theories.
- ▶ The basic paradigm of economics: to understand the functioning of societies as an adaptation to the underlying principle of scarcity.
- ▶ Why economics considers itself a methodology, and not a field of application.
- ▶ Why the opposite of 'positive' is **no longer** 'negative', but 'normative'.
- ▶ The relevance of opportunity costs.
- ▶ How to think and make decisions like an economist.

Literature

- ▶ Kolmar (2022), *Principles of Microeconomics*, Ch. 1.
- ▶ Kolmar/Hoffmann (2022), *Workbook for Principles of Microeconomics*, Ch. 1.



What Do Economists Do?

- ▶ Why are some people/countries rich and others poor?
- ▶ Why is it profitable for internet platforms to charge you a price of zero?
↳ advertisement
- ▶ Can majority voting tell us what “the people” want?

“Economics reaches far beyond markets: voting systems, military and political conflicts, evolutionary biology, auctions, internet platforms, law, philosophy . . .”

Kolmar (2022), *Principles of Microeconomics*, p. 4



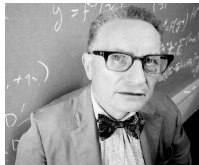
What do all these questions have in common?

“Economics is the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses.”

Robbins (1932), *An Essay on the Nature and Significance of Economic Science*, p. 15



LIONEL C.
ROBBINS
1898–1984



PAUL A.
SAMUELSON
1915–2009

“Economics is the study of how men and society choose, with or without the use of money, to employ scarce productive resources which could have alternative uses, to produce various commodities over time and distribute them for consumption, now and in the future among various people and groups of society.”

Samuelson (1948), *Economics: An Introductory Analysis*



Scarcity (1/2)

"Economics is the study of how society manages its scarce resources."

Mankiw (2021), *Principles of Economics*, Ch. 1



N. GREGORY
MANKIW
1958*

Scarcity

- ▶ Economics looks at the world from the perspective of scarcity.
- ▶ Scarcity exists if wants exceed the available means to satisfy them.



“Economics is the study of how individuals and societies manage goods and resources, which can be objectively as well as subjectively scarce.”

Kolmar (2022), *Principles of Microeconomics*, p. 4

Methodology

Therefore, economics is a methodology, not a field of application.



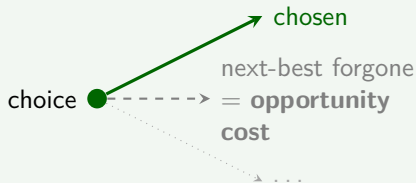
Opportunity Costs

*“The value one connects to the **next-best alternative forgone** is the **opportunity cost** of one’s choice.”*

Kolmar (2022), *Principles of Microeconomics*, p. 5

Interpretation

Every choice means giving up alternatives: if a person wants more of something, he or she must give up something else. What the choice truly costs is the value of the **next-best alternative forgone**, not the money spent.



Two Ways of Responding to Scarcity

- ▶ For given ends, expand the means (economics).
- ▶ For given means, work on your ends (philosophy, e.g., Buddhism).

Kolmar (2022), *Principles of Microeconomics*, Digression 1.1, p. 5



Positive vs. Normative Theories

Two perspectives for analyzing the impact of scarcity on society:

Positive Theories

- ▶ How **do** people deal with the phenomenon of scarcity?
- ▶ Statements about 'what is' (= **descriptive**).

Normative Theories

- ▶ How **should** people deal with the phenomenon of scarcity?
- ▶ Prescriptions about 'what ought to be' (= **prescriptive**); they rely on a **value judgment**.



Economics develops theories that help us better understand the world; better decisions and solved problems are a (welcome) **byproduct**.

Theory

A relatively broad conceptual approach that makes reasonable conjectures about causal relationships in the world.

Reasonable

A conjecture is reasonable if it is *consensual among experts* (in a non-hierarchical discourse).

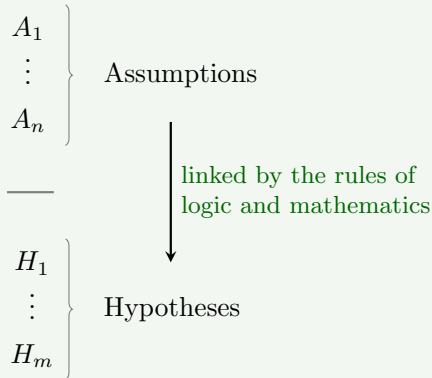


Model

*“A model is a **collection of assumptions and hypotheses** that are linked by the rules of logic and mathematics.”*

Kolmar (2022), *Principles of Microeconomics*,
p. 14

Structure of a Model



Three Types of Propositions (1/4)

1. Analytical Propositions

- ▶ Example: “ $2+2=4$ ”
- ▶ They can be correct or incorrect according to internal standards.
- ▶ Epistemic standard: logical correctness (models are complex analytical propositions).



Three Types of Propositions (2/4)

2. Synthetic Propositions

- ▶ Example #1: "*this bottle is blue*" (singular proposition)
- ▶ Example #2: "*all swans are white*" (universal proposition)
- ▶ They can be correct or incorrect according to external standards.
- ▶ Epistemic standard: empirical correctness ("*this bottle is in fact blue*").



Three Types of Propositions (3/4)

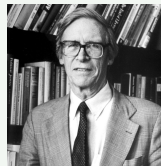
3. Normative Propositions

- ▶ Example #1: “*this coffee tastes good*”
- ▶ Example #2: “*equality is just*”
- ▶ They cannot be true or false; they can be justified or criticized according to some specific evaluative standard.

Rawls' Veil of Ignorance

- ▶ Choose the principles of justice as if you did not know your own position in society, behind a “veil of ignorance,” so the norms cannot be biased in your own favor.
- ▶ An evaluative standard against which normative propositions can be justified.

Rawls (1971), *A Theory of Justice*



JOHN RAWLS
1921–2002



Three Types of Propositions (4/4)

Overview

Type	Example	Epistemic standard	Truth value?
Analytical	" $2+2=4$ "	logical correctness	yes
Synthetic	"this bottle is blue"	empirical correctness	yes
Normative	"equality is just"	justifiability	no

Note on the textbook

Kolmar (2022, Sec. 1.2.7) only distinguishes two kinds of statements:

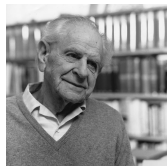
- ▶ **positive (descriptive)**
- ▶ **normative**

The finer analytical/synthetic split is added here on the slides.



The standard epistemological approach in economics

- ▶ Scientific theories can never be *verified*, but can, in principle, be *falsified*.
- ▶ Falsifiable: there are empirically accessible conditions under which the theory would fail.
- ▶ Good theories rely on logically consistent models.
- ▶ Good theories are frugal (Ockham's razor).



KARL R.
POPPER
1902–1994

A Good Theory

A good theory is logically consistent, frugal, and has a large empirical content: falsifiable, but not (yet) falsified.



Three Levels of Analysis (1/3)

Economics traditionally distinguishes three levels of analysis when studying societal phenomena:

Individual level

How do people behave under scarcity? (*movies or dinner?*)

Interaction level

How do people interact? (*A goes to the movies; B wants to meet A . . .*)

Aggregate level

How does the economy function as a whole? (*growth, unemployment*)

Microeconomics

Macroeconomics



Three Levels of Analysis (2/3)

Methodological Individualism

Social phenomena must be explained by reference to *individual* behavior: aggregate regularities are, in principle, traceable back to patterns of behavior on the individual and interaction levels.



Three Levels of Analysis (3/3)

Microeconomics

“Microeconomists study individual decision-making at the individual and interaction levels . . .”

Kolmar (2022), *Principles of Microeconomics*, p. 10



Modern macroeconomics is largely based on microeconomic principles.

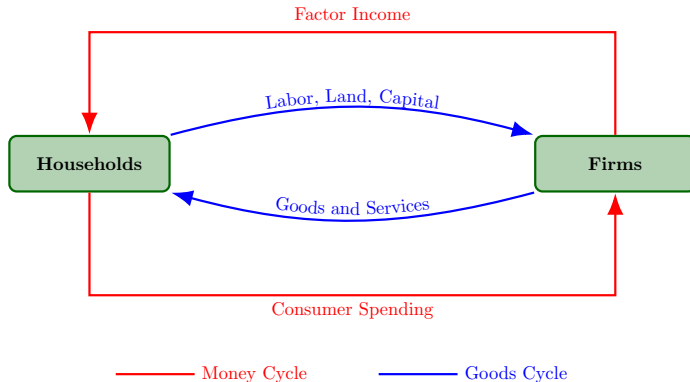
Macroeconomics

“... whereas macroeconomists study economic phenomena at the aggregate level.”

Kolmar (2022), *Principles of Microeconomics*, p. 10



Example of an Economic Model: Circular Flow Diagram



What does this model isolate?

The exchange between households and firms in goods and factor markets, abstracting from everything else (state, banks, the rest of the world).

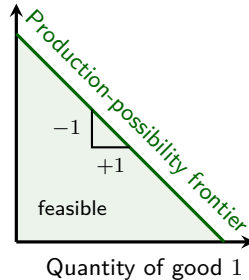


Example of an Economic Model: Production-Possibility Frontier



Andreas Gursky (1999), *Chicago Board of Trade II*

Quantity of good 2



What does this model isolate?

The production trade-off itself: more of one good means giving up units of the other; the (constant) slope is the opportunity cost. It abstracts from everything else: more than two goods, other shapes of the PPF.



Summary: What You Should Take Away

- ▶ Scarcity exists if wants exceed the available means; it is the reason economics exists.
- ▶ Every decision therefore has an **opportunity cost**: the value of the next-best alternative forgone.
- ▶ **Positive** theories ask how people *do* deal with scarcity; **normative** theories ask how they *should* (relying on a value judgment).
- ▶ Theories are reasonable conjectures about the world; a **model** is a collection of assumptions and hypotheses linked by logic and mathematics.
- ▶ A good theory is logically consistent, frugal, and has a large empirical content: falsifiable, but not (yet) falsified (Popper's Critical Rationalism).
- ▶ Economics analyzes scarcity on three levels: **individual**, **interaction**, and **aggregate**; microeconomics covers the first two.



Original Exam Questions

Decide whether the following statements are *true* or *false*.



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